



BIAS IN NETWORKS: Friendship Paradox & Perception Bias

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DSCI 531

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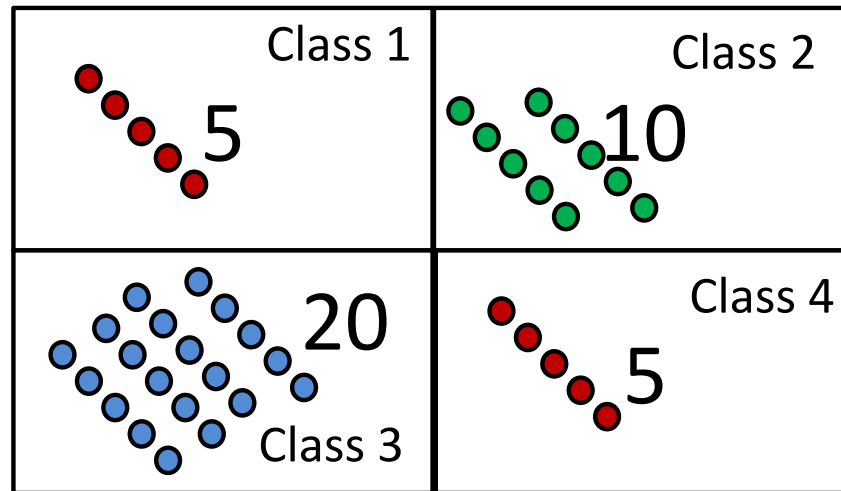
Class size paradox

- You arrive at a college that boasts average class size of 10 students
- To confirm, you survey students at random about how large their class is
 - 5
 - 5
 - 20
 - 10
 - 5
 - 20
 - ...
 - What is the average class size



Class size paradox

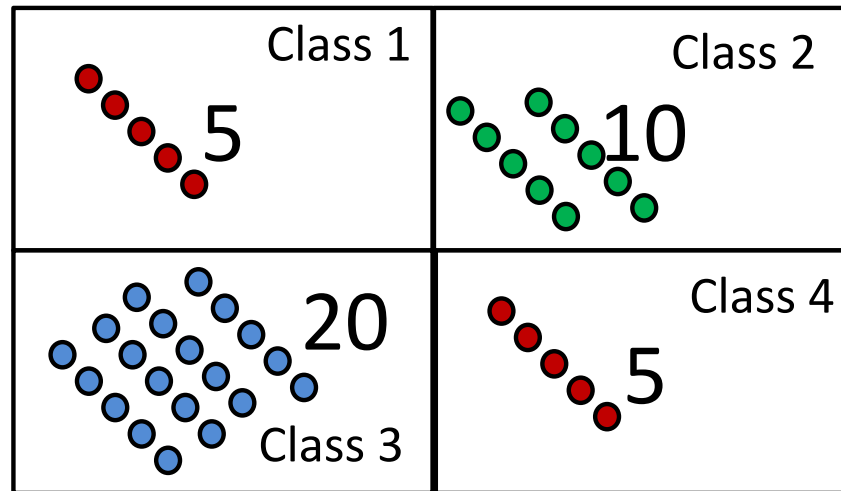
- Average class size is ...





Class size paradox

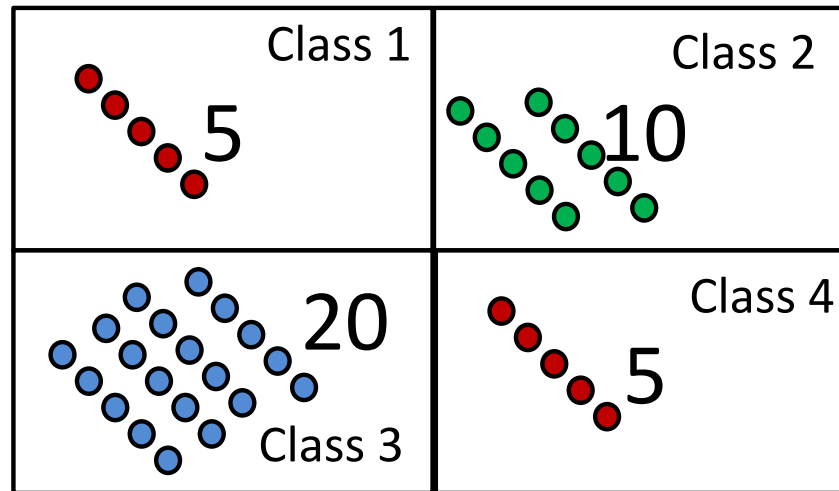
- Size of class a student experiences on average is ...
 - (average number of classmates)





Class size paradox

- Average person is associated with larger events than the size of the average event.

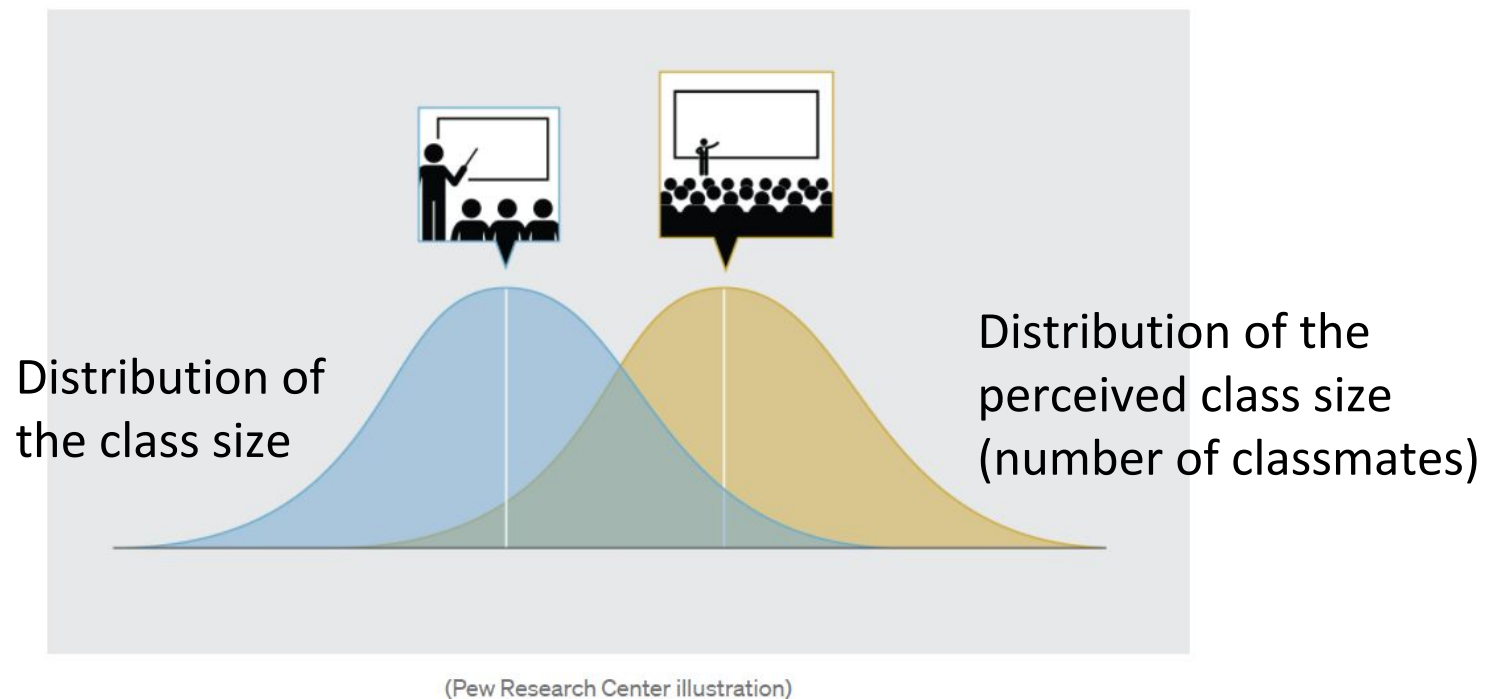


Average size of class is 10

Size of class an average student experiences is 12.75



Class size paradox



Households are larger at the individual level than at the group level



Average person's household is larger than average household

Household size, by level of analysis

	Average individual	Average household
Sub-Saharan Africa	6.9	5.0
Middle East-North Africa	6.2	5.1
Asia-Pacific	5.0	4.0
Latin America-Caribbean	4.6	3.6
North America	3.3	2.5
Europe	3.1	2.4
World	4.9	3.9

Source: Pew Research Center analysis of 2010-2018 census and survey data. See [Methodology for details](#)

- Discrepancies between the average household size and the average experienced by individuals tend to be largest in countries where typical households are large
- The difference between household size as experienced by the average person in the world (4.9 members) is one whole person more than the size of the average household (3.9)



Differences by household type

Household types: Distributions change when perspective shifts

% in each household type by level of analysis, all countries combined

	Asia-Pacific	Europe	Individuals			Sub-Saharan Africa	World
			Latin America-Caribbean	Middle East-North Africa	North America		
Extended	45%	26%	32%	27%	11%	35%	38%
Two-parent	31	26	39	56	33	37	33
Couple	7	19	6	3	20	2	8
Adult child	10	9	10	9	14	2	9
Solo	3	13	3	1	11	2	4
Single-parent	2	4	5	2	9	6	4
Polygamous	<0.5	<0.5	<0.5	0.9	<0.5	11	2
Other	1	2	4	<0.5	1	4	2
	Asia-Pacific	Europe	Households			Sub-Saharan Africa	World
			Latin America-Caribbean	Middle East-North Africa	North America		
Extended	32%	17%	23%	19%	8%	29%	26%
Two-parent	28	16	33	53	19	34	27
Couple	14	23	11	8	26	5	15
Adult child	12	8	12	11	12	3	10
Solo	11	30	11	5	28	12	15
Single-parent	3	4	6	3	7	8	4
Polygamous	<0.5	<0.5	<0.5	<0.5	<0.5	5	0.5
Other	0.9	2	3	<0.5	1	3	1

Source: Pew Research Center analysis of 2010-2018 census and survey data. See Methodology for details.
*Religion and Living Arrangements Around the World"

PEW RESEARCH CENTER

- Differences in the shares of people who live in different household *types*.
 - For example, 26% of households in the 130 countries contained extended families, but 38% of people in the same countries live in extended family households, making this the most common living arrangement in the world.
 - Solo households make up 15% of homes globally, but only 4% of people live alone.
- Similar arguments for white people experience shops to be more crowded than they are, and freeways to be more congested than they really are

This explains why your perceptions are biased



Your friends
are happier



Your friends
drink more





The problem of binge drinking among (US) students

- 10-25% of college students are “heavy” or “problem” drinkers.

This causes issues:

- collegiate adjustment problems
- accidents
- academic failure
- vandalism
- development of mental health disorders

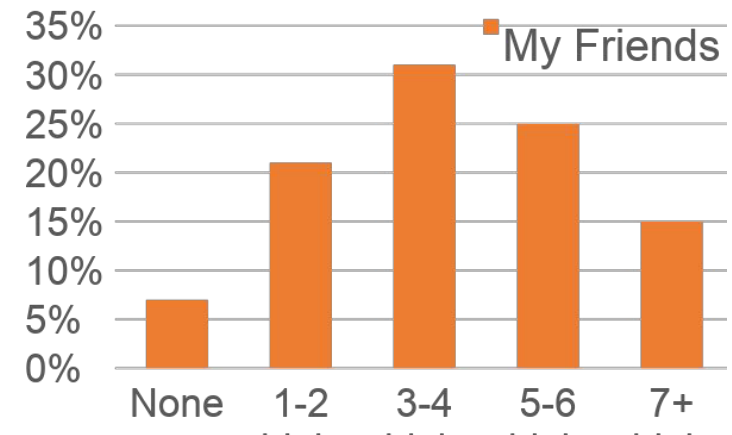
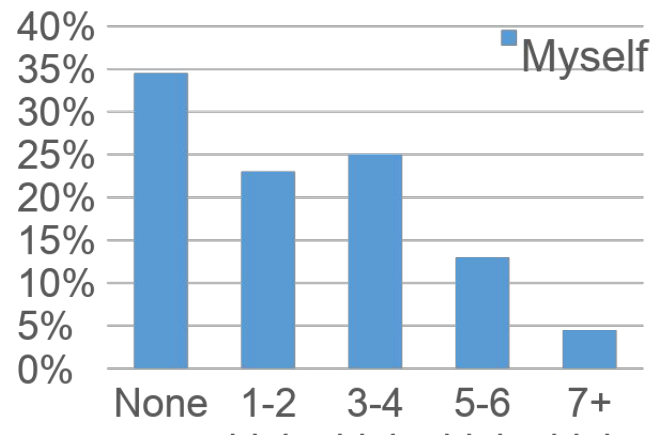


Biases in the Perception of Drinking Norms among College Students*

JOHN S. BAER, PH.D., ALAN STACY, PH.D., AND MARY LARIMER, PH.D.

Department of Psychology, NI-25, Addictive Behaviors Research Center, University of Washington, Seattle, Washington 98195

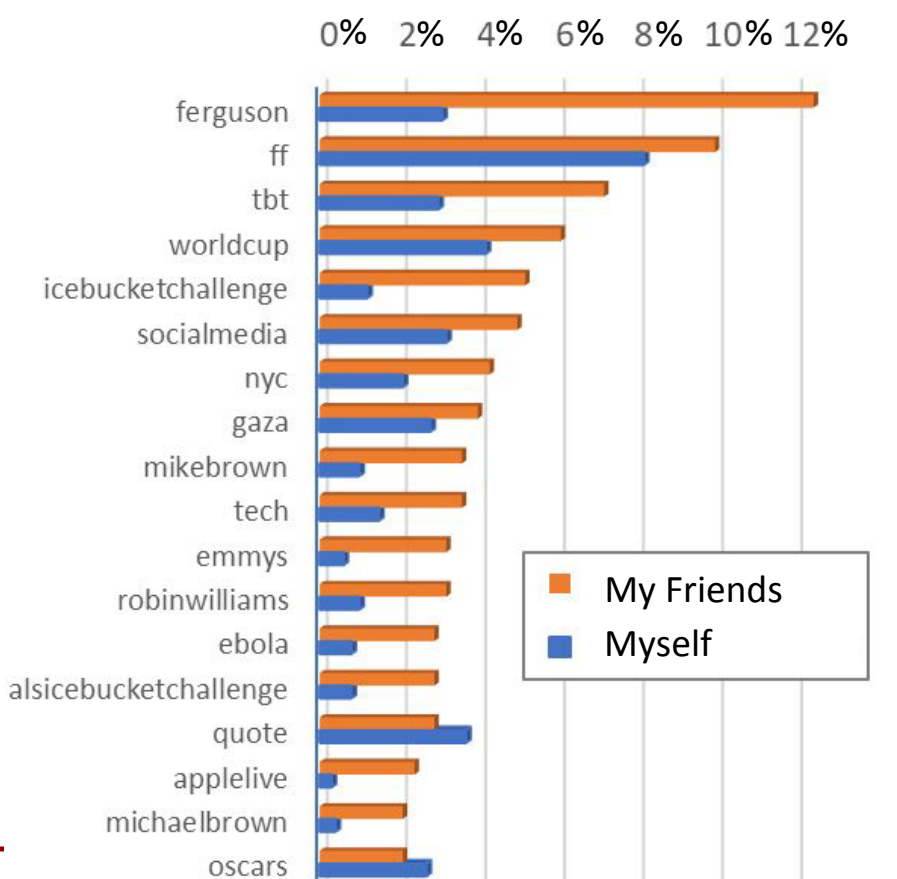
How many alcoholic drinks are consumed at a party?
Virtually all students report that their friends drink more



Source: Most Students Do PartySafe@Cal



Twitter users overestimate the popularity of some topics



- Compare global to local popularity of topics
 - Myself = Global popularity = fraction of all people talking about a topic
 - My Friends = Local popularity = fraction of friends talking about the topic
- Topics: Popular Twitter hashtags in Summer 2014
- Current events topics appear more popular than they really are

[Alipourfard et al (2020) "Friendship paradox biases perceptions in directed networks", *Nature Communications*]



Why?

The Social-Network Illusion That Tricks Your Mind

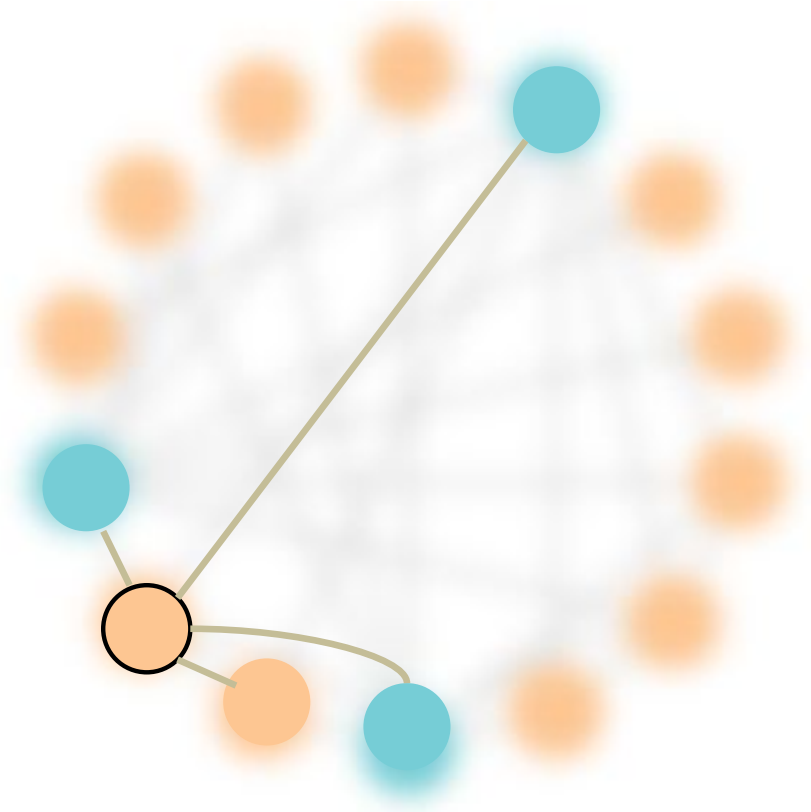
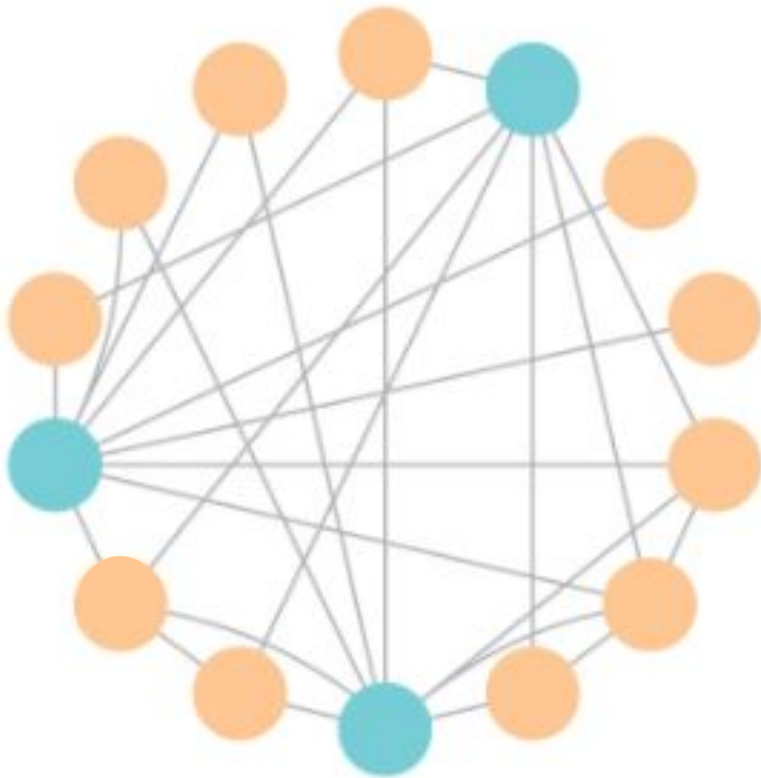
Network scientists have discovered how social networks can create the illusion that something is common when it is actually rare.

June 30, 2015

- Friends are not a random sample of the population!
- People seldom see representative sample of the population.
- Instead, their samples — their friends — are highly biased samples of the population
- This, in turn, biases their perception of how common different attributes are in the general population
- These over- and underestimation errors, which we call **perception biases**, affect people's judgements of minority- and majority-group sizes .



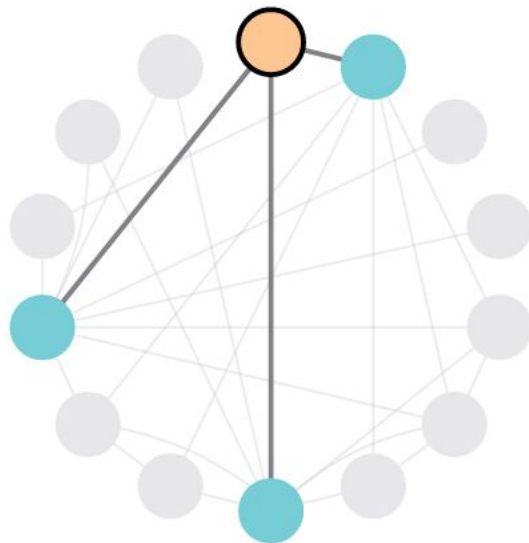
Social perception bias is a network effect



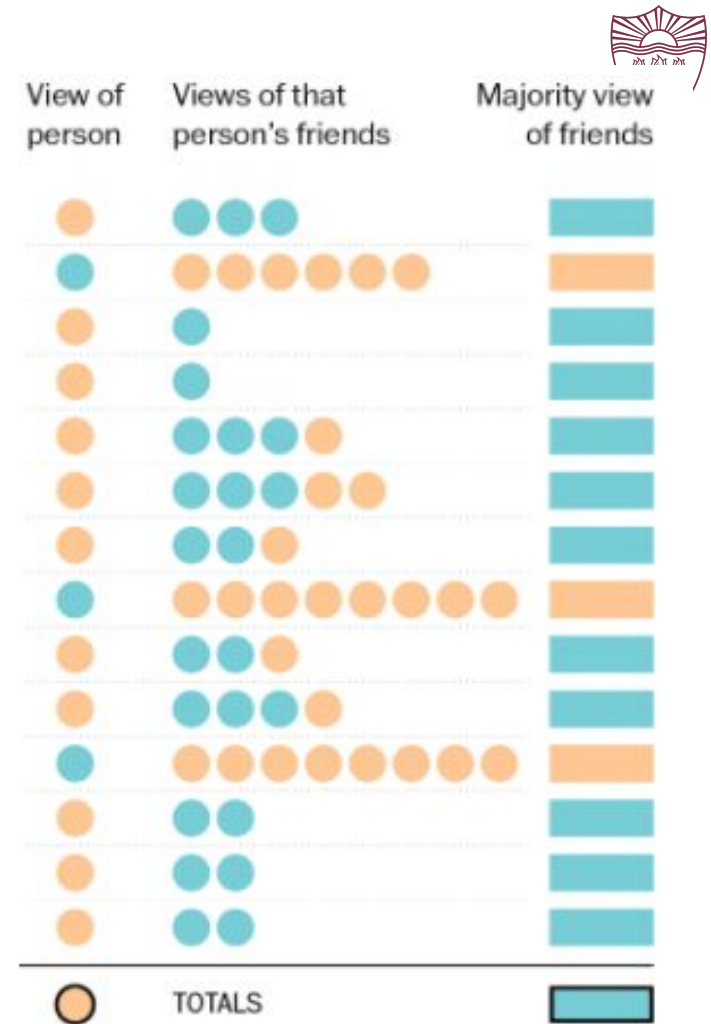
Information Sciences Institute
[Lerman, Wu & Yan (2016) The “Majority Illusion” in Social Networks, in *Plos One*.]

Local vs global views

100% of this person's friends think baseball caps are trendy.



K. Schaul, "A quick puzzle to tell whether you know what people are thinking", *Wonkblog, Washington Post*
<https://www.washingtonpost.com/graphics/business/wonkblog/majority-illusion/>



FRIENDSHIP PARADOX IN NETWORKS





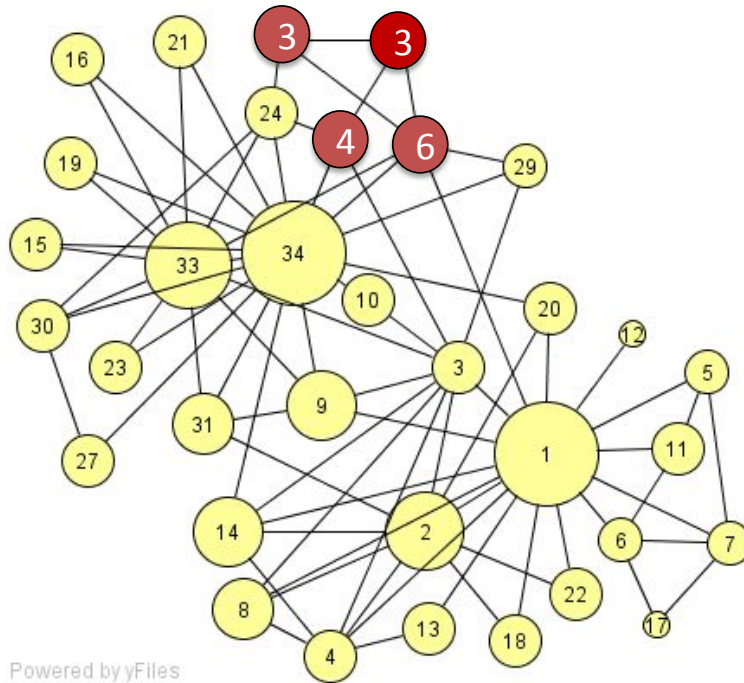
Friendship paradox

Your friends have more friends than you do, on average [Feld, 1991]



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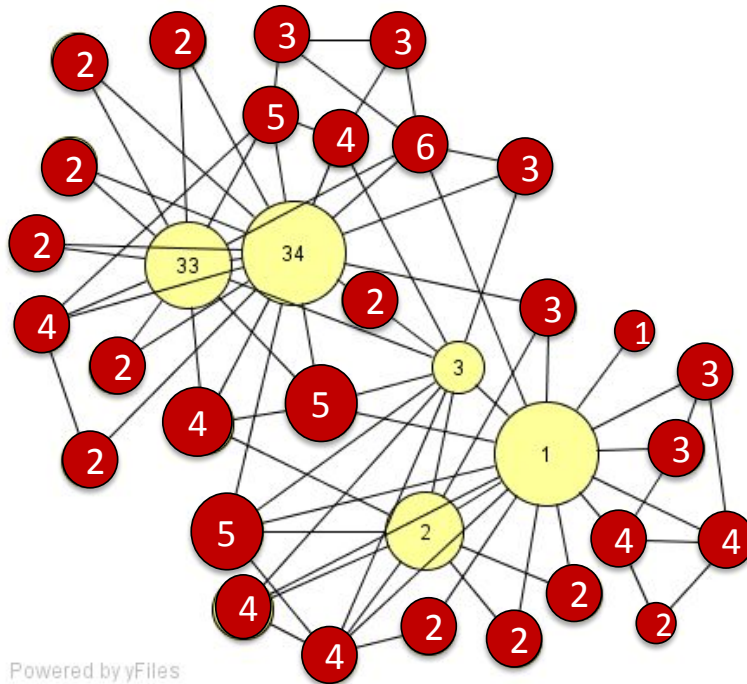


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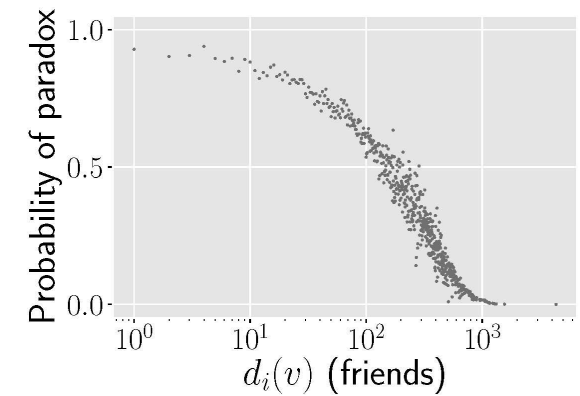


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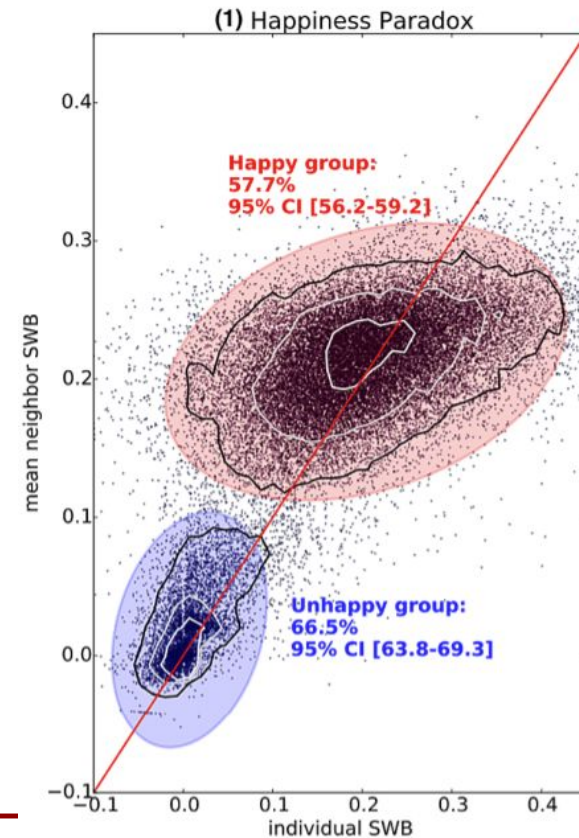
Generalized friendship paradox

Your friends are more X than you are, on average [Hodas et al., 2013, Eom & Jo, 2014]

- Your friends are happier than you are, on average (Bollen 2017)
- Your coauthors are more productive and more recognized than you are, on average (Eom & Jo 2014)
- Your coauthors have more impact than you do, on average (Benevenuto 2016)
- True whenever more popular people have more X

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$$F = \langle x \rangle_{nn} - \langle x \rangle = \frac{\rho_{kx} \sigma_k \sigma_x}{\langle k \rangle}.$$





Friendship paradox

Your friends have more friends than you do, on average [Feld, 1991]



Generalized friendship paradox

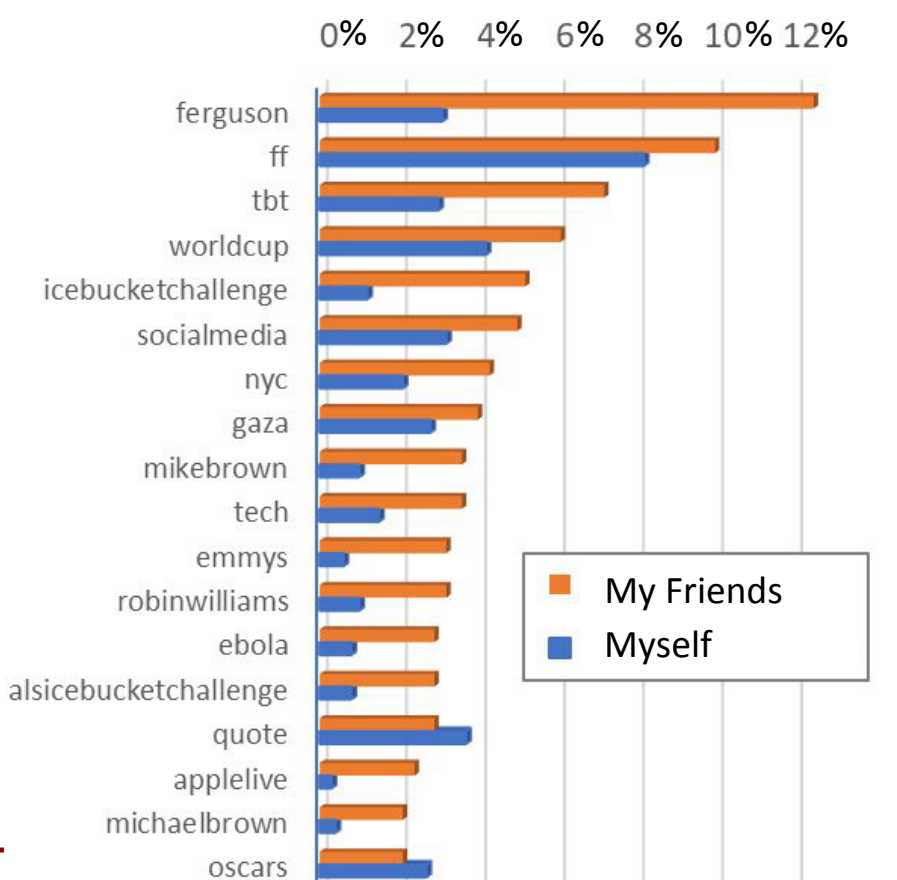
Your friends are more X than you are, on average [Hodas et al., 2013, Eom & Jo, 2014]



Perception bias

Some traits appear more popular within social circles than they really are [Alipourfard et al, 2020]

Perception bias on Twitter: users overestimate the popularity of some topics

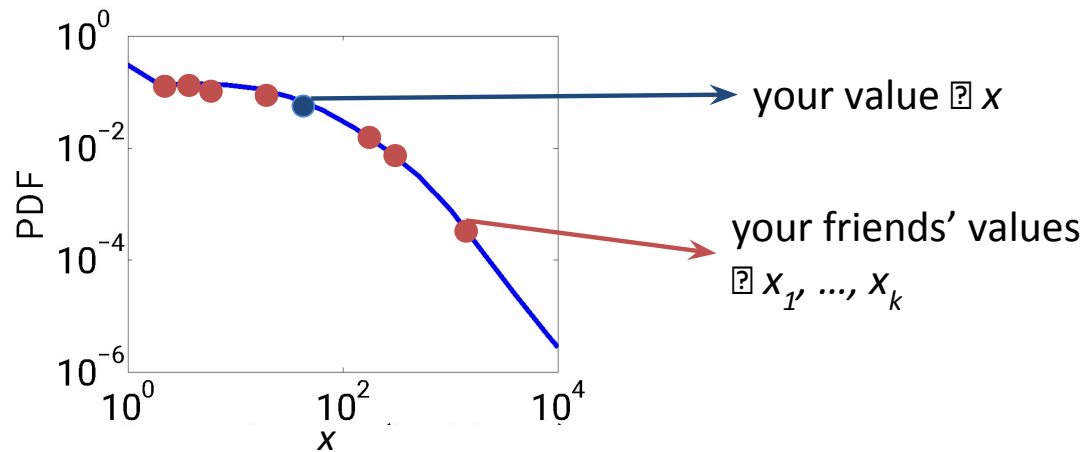


- Compare global to local popularity of topics
 - Myself = Global popularity = fraction of all people talking about a topic
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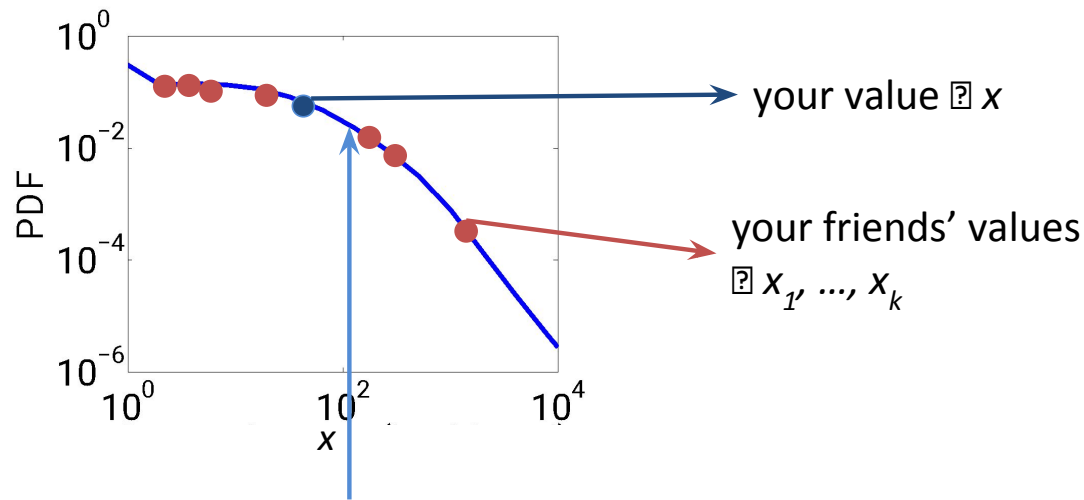


Friendship paradox as a byproduct of sampling from a heterogeneous distribution





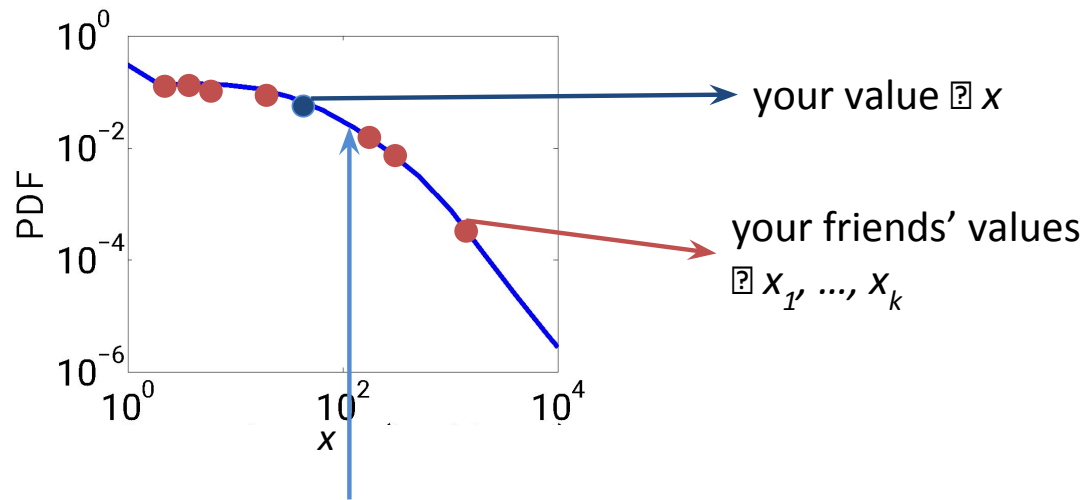
Friendship paradox as a byproduct of sampling from a heterogeneous distribution



Sample mean grows with k



Friendship paradox as a byproduct of sampling from a heterogeneous distribution



☐ **use the median instead**



Friendship paradox

Your friends have more friends than you do, on average [Feld, 1991]

Strong friendship paradox

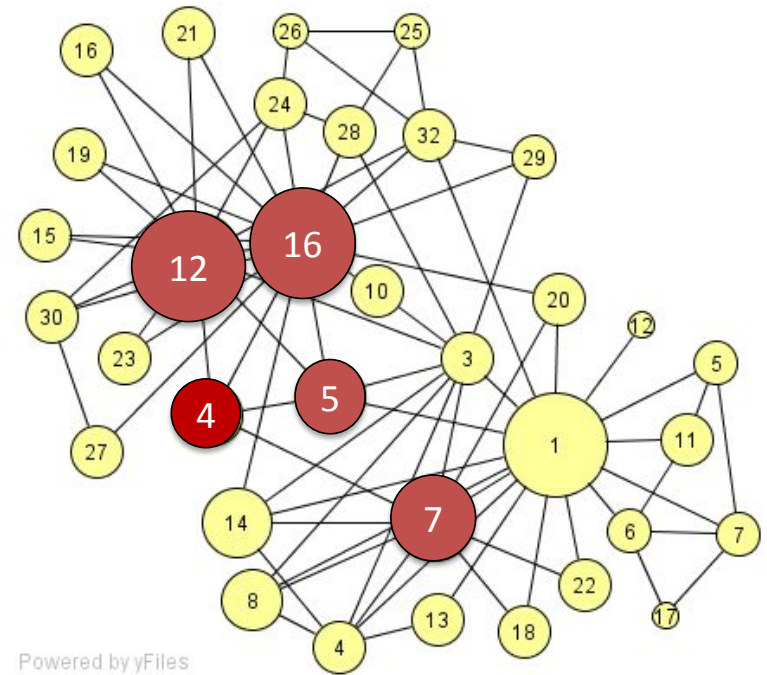
Most of your friends have more friends than you do [Kooti et al., 2014]



Strong friendship paradox

Most of your friends have more friends than you do

[Kooti, Hodas and Lerman, 2014].



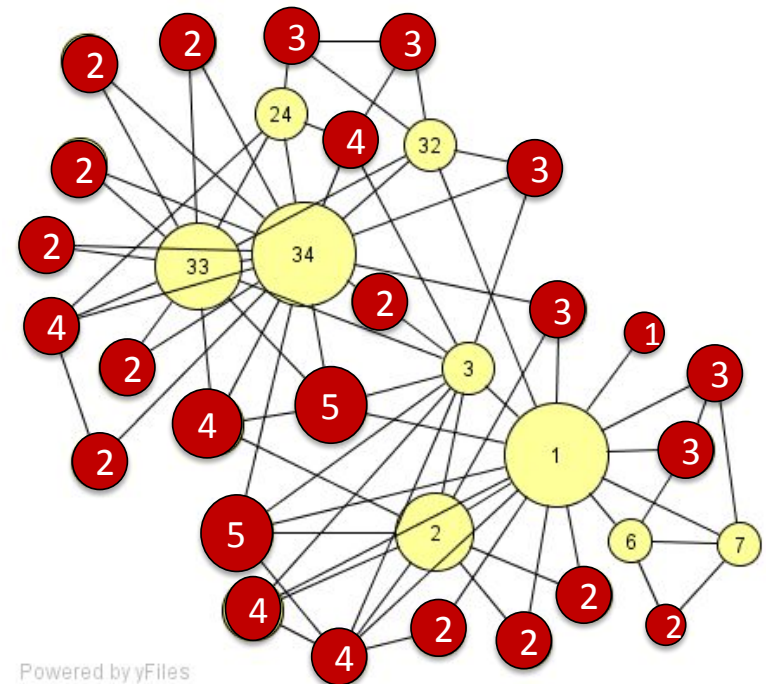
Powered by yFiles



Strong friendship paradox

Most of your friends have more friends than you do

[Kooti, Hodas and Lerman, 2014].





Friendship paradox is very common...

Network	Type	Nodes	Probability of paradox
LiveJournal	Social	3,997,962	84%
Twitter	Social	780,000	98%
Skitter	Internet	1,696,415	89%
Google	Hyperlink	875,713	77%
ProsperLoan	Social Finance	89,269	88%
ArXiv	Citation	34,546	79%
WordNet	Semantic	146,005	75%

But wait, there is more...

Friendship paradox

Your friends have more friends than you do, on average [Feld, 1991]



Generalized friendship paradox

Your friends are more X than you are, on average [Hodas et al., 2013, Eom & Jo, 2014]



Perception bias

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Generalized strong friendship

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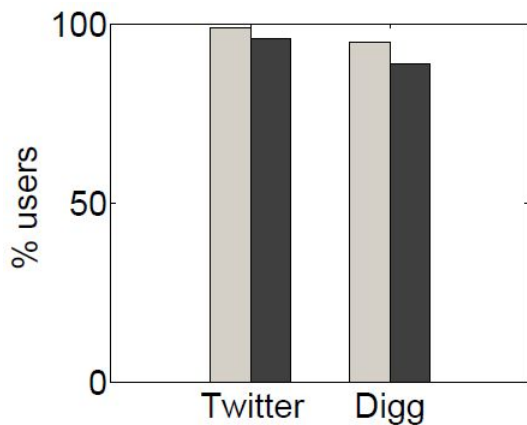


Majority illusion

Most of your friends have a trait, even when it is rare. [Lerman et al, 2016]

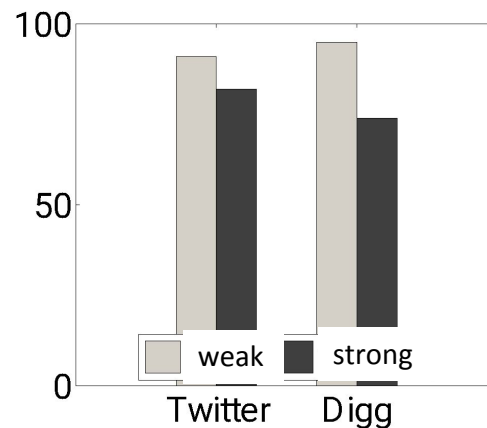
Strong Friendship Paradox

Most of your friends (& followers) are more popular than you are



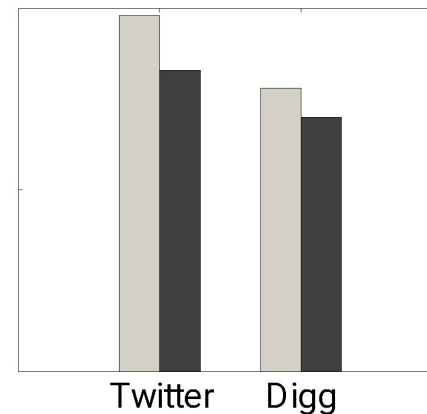
Strong Activity Paradox

Most of your friends tweet more than you do



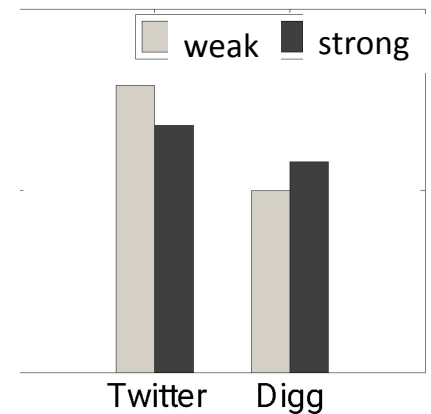
Strong Diversity Paradox

Most of your friends see more diverse content than you do



Strong Virality Paradox

Most of your friends see more viral content than you do



[Kooti, et al (2014) "Network Weirdness: Exploring the origins of network paradoxes" in ICWSM]

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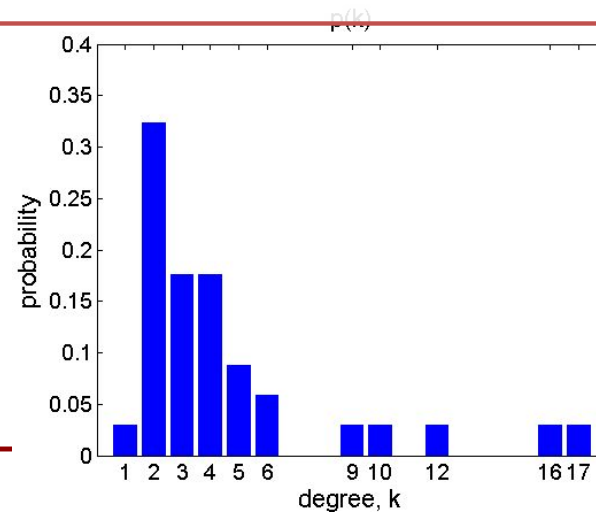
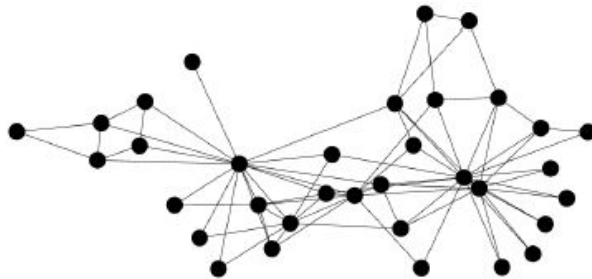
QUANTIFYING FRIENDSHIP PARADOX: THE ROLE OF NETWORK STRUCTURE



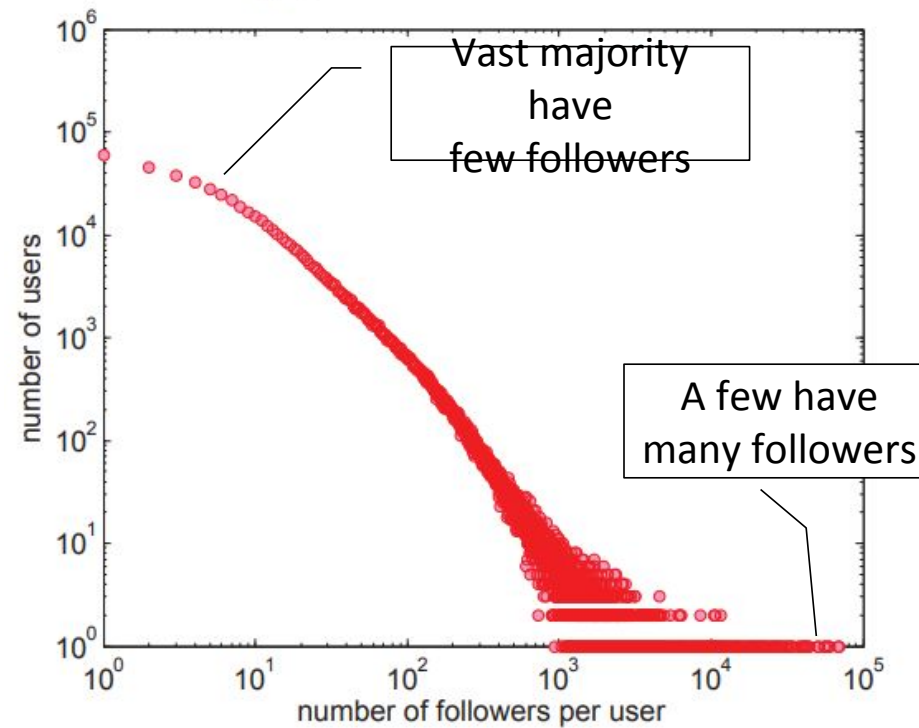


Degree distribution

- Node degree distribution $p(k)$
 - Probability that a randomly selected node has degree k .
- Explains Feld's friendship paradox
 - Any skewed degree distribution (variance > 0) will lead to a paradox



Distribution of popularity on Twitter (ca 2010)





Assortativity



... Nodes do not link at random

- Joint degree distribution of connected *pairs of nodes* $e(k, k')$
 - Probability that a randomly selected edge links nodes with degrees k and k'
 - Explains Feld's friendship paradox, but not strong one

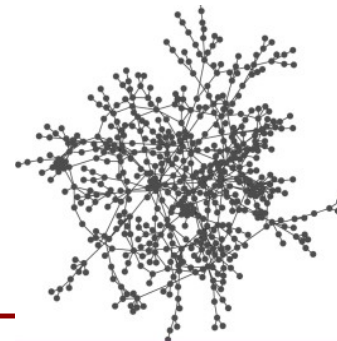
- **Degree assortativity**

MEJ Newman, Assortative Mixing in Networks, *Phys Rev Lett* 89 208701 (2002)

disassortative



assortative





Transsortativity

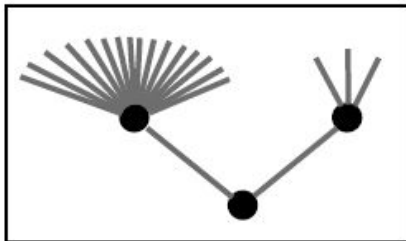


... nodes do not link at random

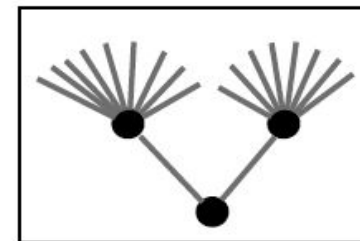
- Joint degree distribution of connected *triplets* of nodes
- **Transsortativity**: neighbors tend to have similar degrees
- Networks can have the same lower-order structure (degree and joint degree distributions) but different higher-order structure

Ngo, Percus, Burghardt & Lerman. The Transsortative Structure of Networks, *in RSPA* (2020)

negative
transsortativity

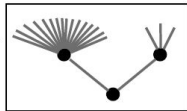


positive
transsortativity

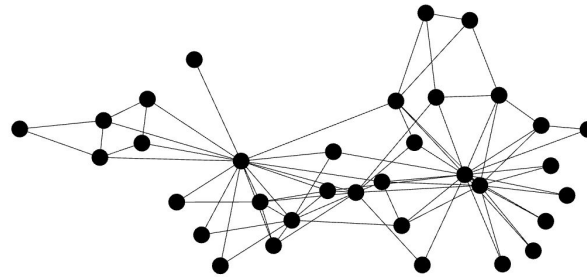




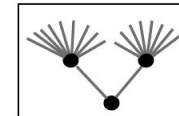
Transsortativity changes network structure



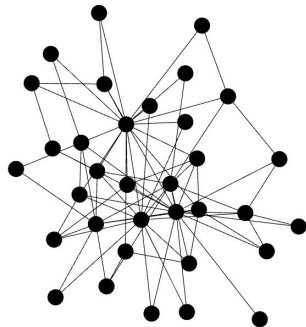
negative
transsortativity



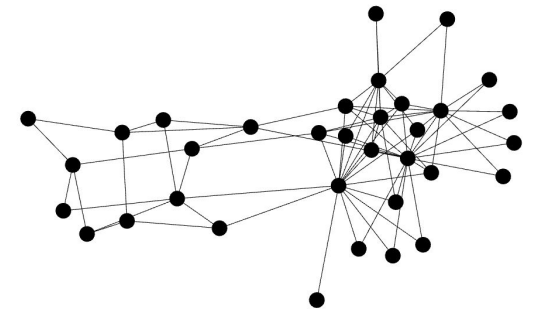
original network



positive
transsortativity



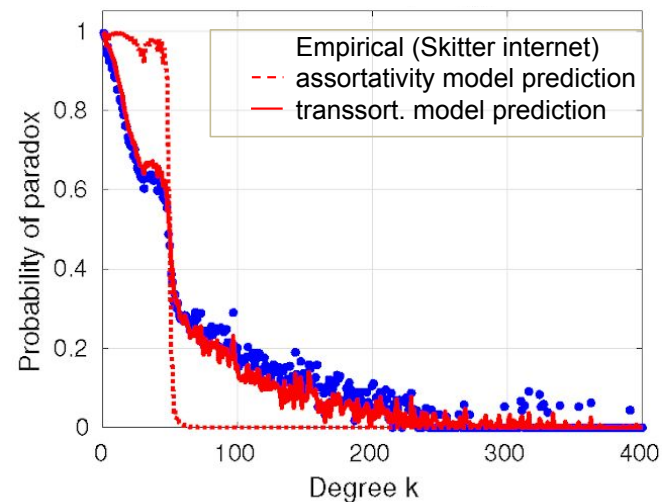
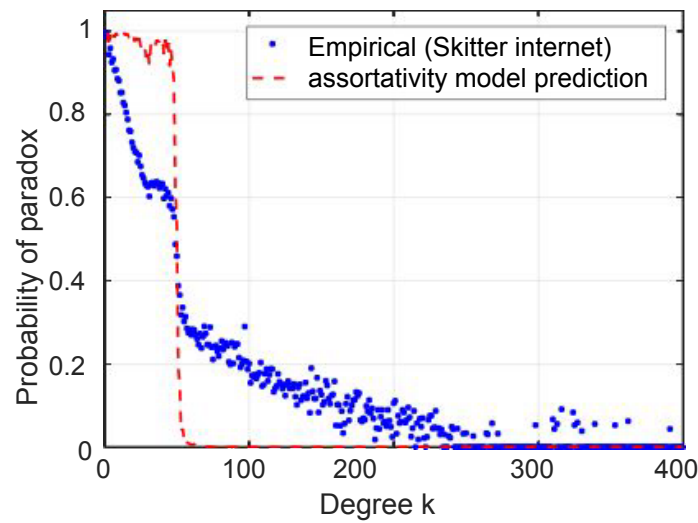
* all networks have the same
lower order structure: degree
and joint degree distributions





Transsortativity explains strong friendship paradox

Given a network, calculate the fraction of degree- k nodes the majority of whose neighbors have a higher degree



* transsortative structure is sufficient to explain the paradox

[Wu et al (2017) "Neighbor-neighbor correlations explain measurement bias in networks" *Scientific Reports*]

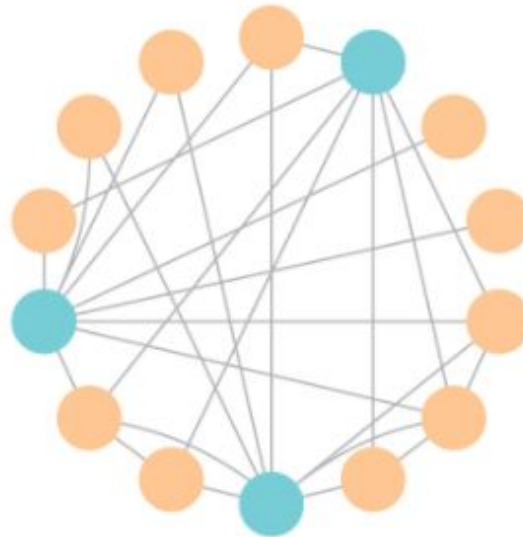
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School of Engineering

From friendship paradox to “majority Illusion”



- Nodes have a trait: active/not, yellow/blue, heavy drinker/teetotaler, ...
- Trait can be rare, but many nodes will overestimate its prevalence

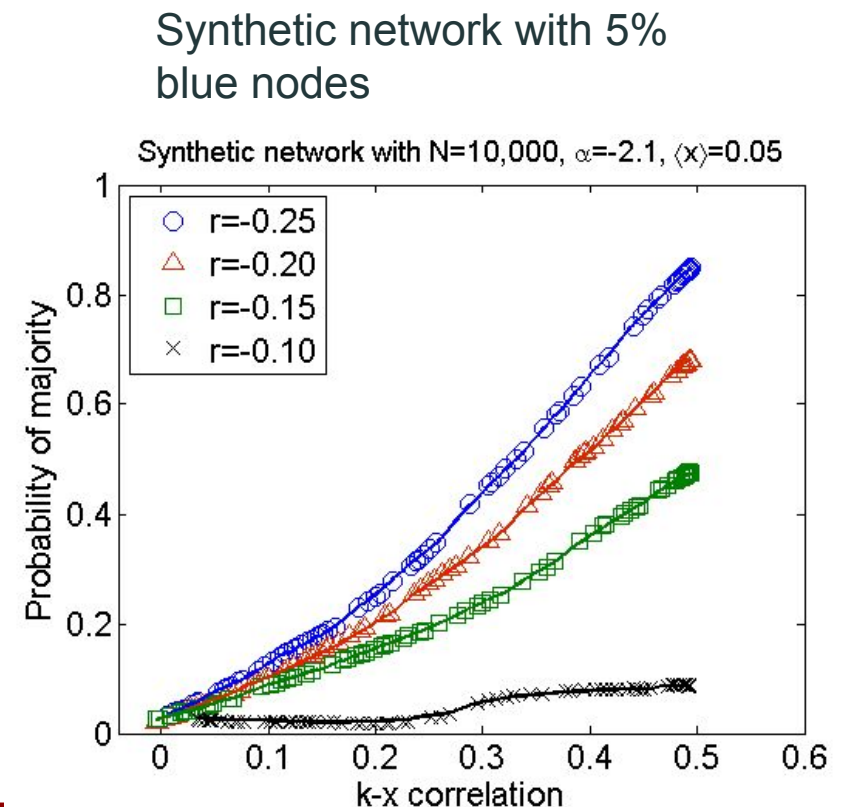


Information Sciences Institute [Lerman, Wu & Yan (2016) The “Majority Illusion” in Social Networks, in *Plos One*.]



Network structure & majority illusion

- Network structure can amplify the majority illusion when:
 - **Influential (high degree) nodes are likely to have the trait:** k - x correlation measures likelihood high-degree nodes have the trait
 - **Skewed degree distribution:** few high-degree nodes amplify the effect
 - **Disassortativity:** High degree nodes link to low degree nodes: negative assortativity ($r_{2k} < 0$)





Inequalities in social networks

Lee, E., Karimi, F., Wagner, C., Jo, H. H., Strohmaier, M., & Galesic, M. (2019). Homophily and minority-group size explain perception biases in social networks. *Nature Human Behaviour*, 1-10.

1.) Homophily influences ranking of minorities

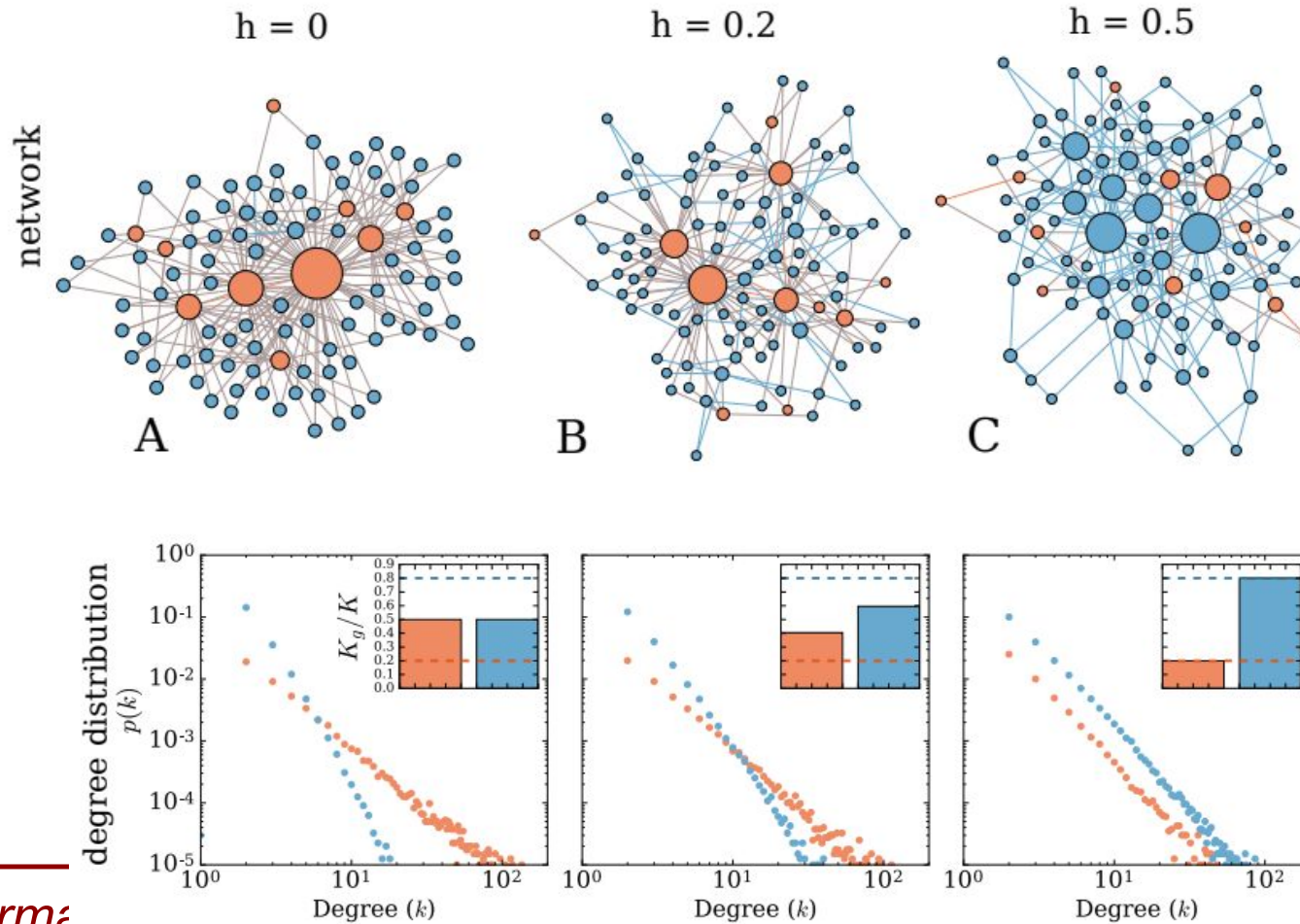
Minorities are more affected by homophilic interactions and their group size than majorities

2.) Homophily can impact perception of groups

Evidence for perception biases on social network and population levels, can be (partially) mitigated

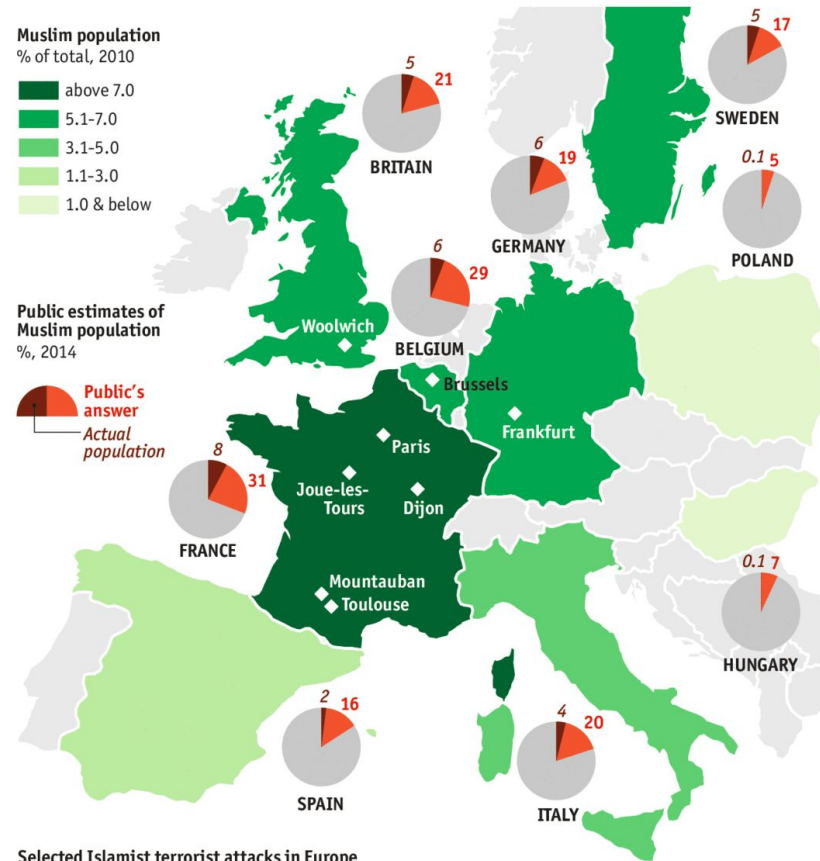


Ranking bias in networks





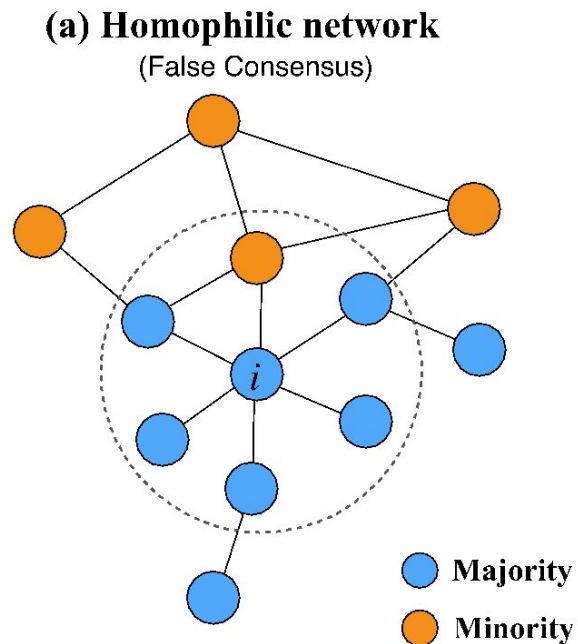
Perception of minorities



Slide courtesy of Markus Strohmaier

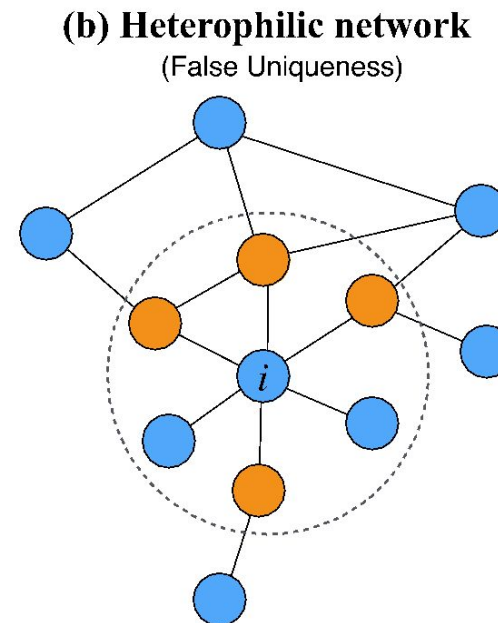


Perception of minorities in social networks



$$P_{\text{indiv}}(i) = 1/6 \approx 16\%$$

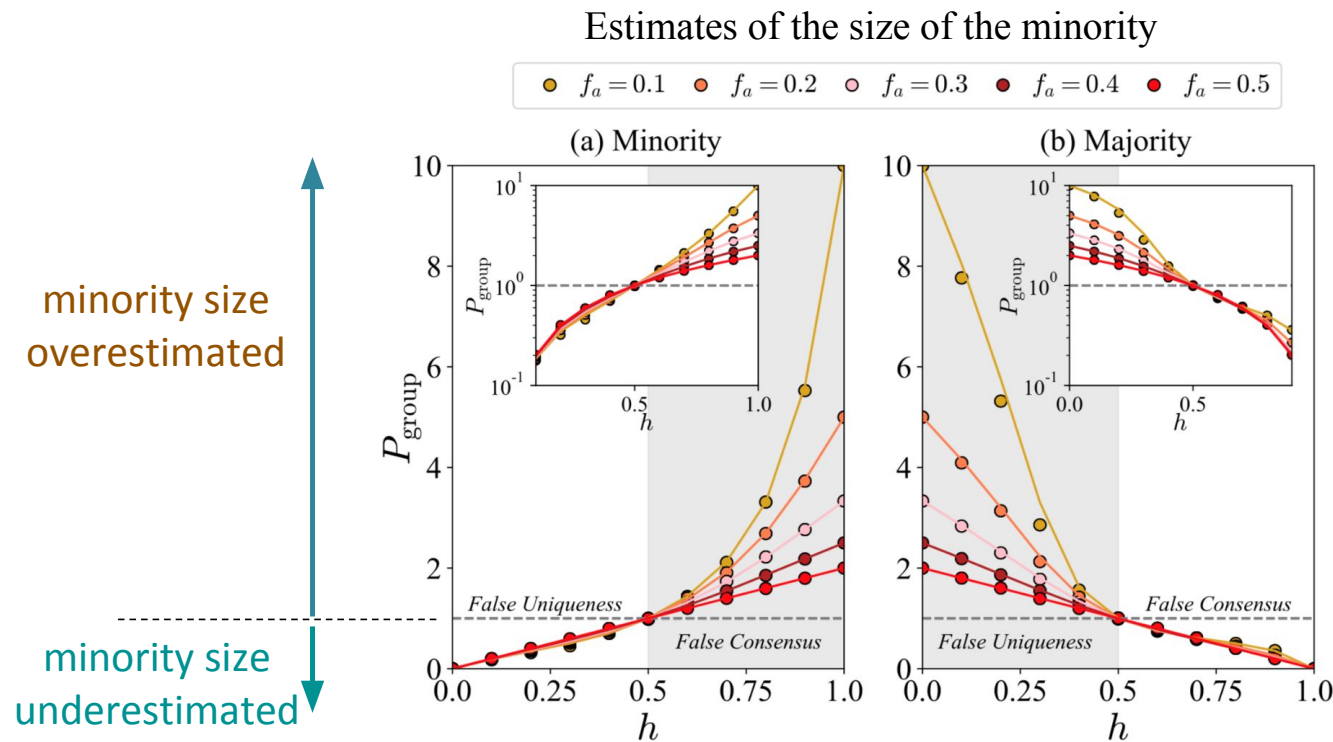
(0.5 fold underestimation)



$$P_{\text{indiv}}(i) = 4/6 \approx 67\%$$

(2 fold overestimation)

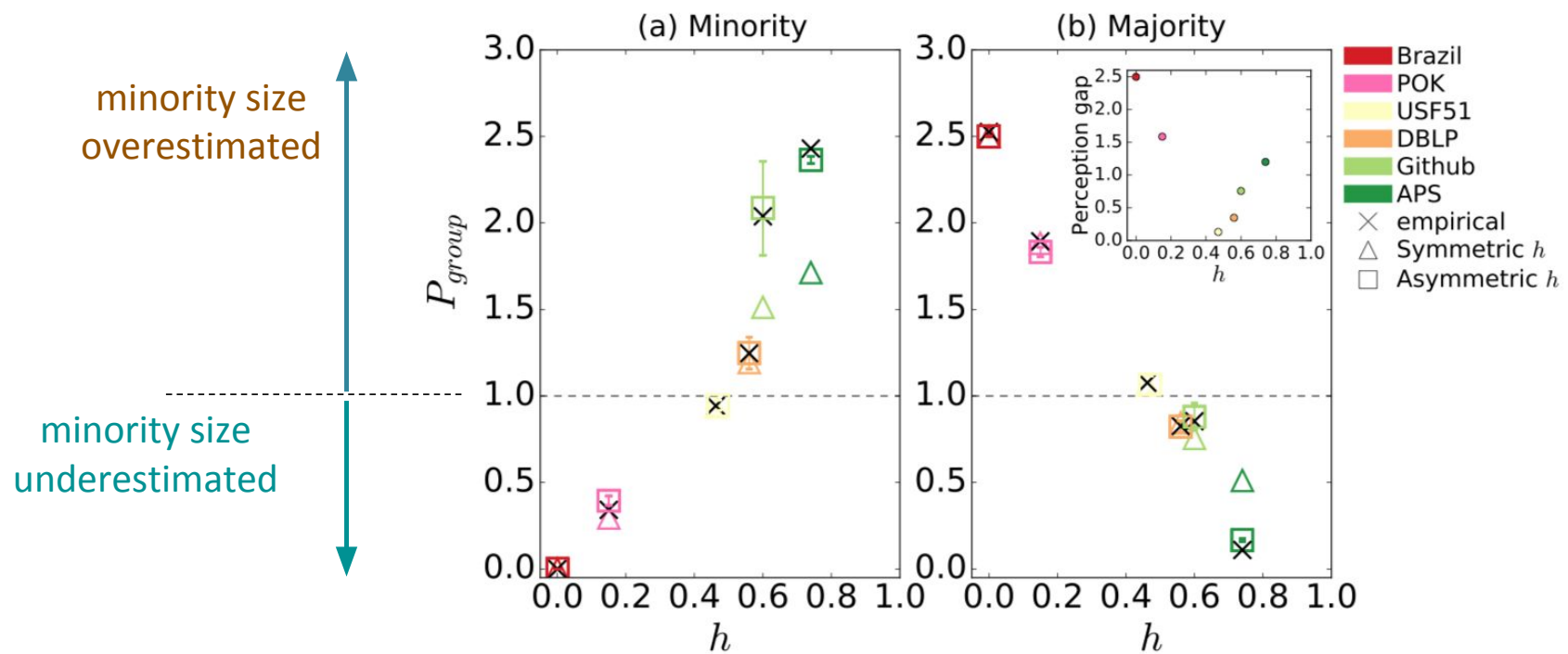
Perception bias vs. homophily and minority size





Perception bias in empirical data

Estimates of the size of the minority

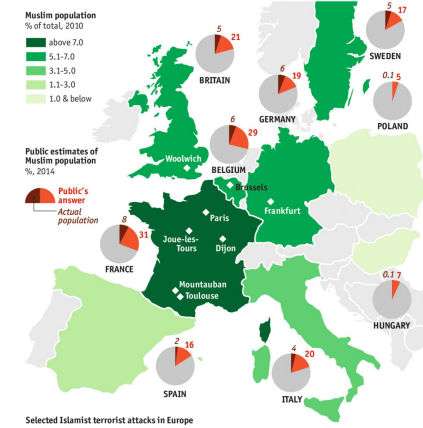


Can we observe similar perception biases on a national level?



- International survey programs including the **US**, **Germany** and **South Korea**:

Characteristic	Question text	Original source of the question text the data for the general population		
		US	Germany	S.Korea
1. Not having money for food	Have there been times in the past 12 months when you did not have enough money to buy food you or your family needed? Yes–No	Gallup World Poll (2010)	Gallup World Poll (2010)	Social Integration Status Survey (2011)
2. Donating to charity	In the past month, have you donated money to a charity? Yes–No	Gallup World Poll (2010)	Gallup World Poll (2010)	Social Survey of Welfare (2017)
3. Experiencing theft	Within the past 12 months, have you had money or property stolen from you or another household member? Yes–No	Gallup World Poll (2010)	Gallup World Poll (2010)	-
4. Religion importance	Is religion an important part of your daily life? Yes–No	Gallup World Poll (2010)	Gallup World Poll (2010)	Religion of Koreans by Gallup (1984-2014)
5. Worship attendance	Have you attended a place of worship or a religious service within the past seven days? Yes–No	Gallup World Poll (2010)	Gallup World Poll (2010)	Religion of Koreans by Gallup (1984-2014)



Can we observe similar perception biases on a national level?



- Examples of different minority / majority issues:

Characteristics	US(%)		Germany(%)		Korea(%)	
	Yes	No	Yes	No	Yes	No
1. Not having money for food	19	81	5	95	3	97
2. Donating to charity	57	43	43	57	26.7	73.7
3. Experiencing theft	12	88	9	91	-	-
4. Religion importance	70	30	27	73	52	48
5. Worship attendance	53	47	33	67	44	56
6. God and morality	47	53	33	67	-	-
7. Belief in a god	64	36	38	62	39	61
8. Smoking	15.2	85	21.9	78	23.9	76.1
9. Military force	76.5	24	50	50	-	-
10. Homosexuality	35.5	64	12.1	87.9	34	58

Table S2. The object ratio of population for US, Germany, and Korea with the survey questions.

+ a crowdworker survey (n=300) asking:

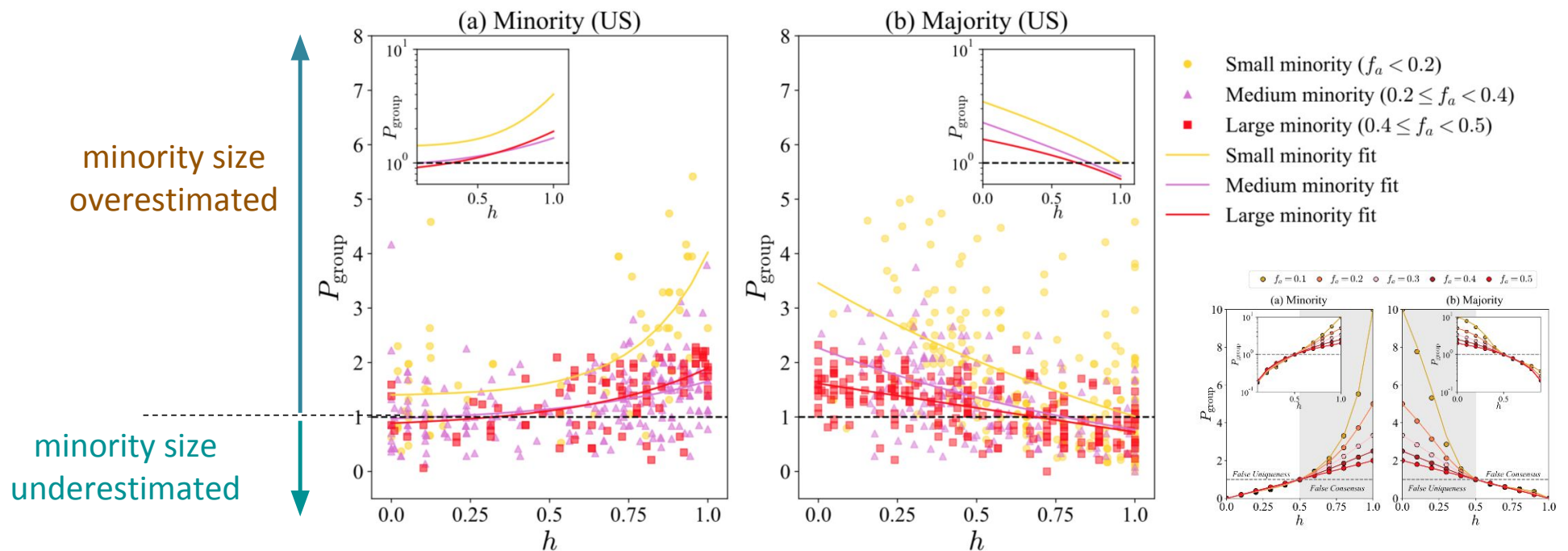
1. Do you have **characteristic x**?
2. How frequent is **characteristic x** in your personal network?
3. How frequent is **characteristic x** in the population of your country?

Info



Perception of minorities on a population level

- Results for minority/majority estimates from the US (results similar for Germany):





Summary

- Friends are not a representative sample of the population
- As a result, your observations/perceptions will be systematically biased
 - E.g., a rare attribute appears far more common than it is
- Impacts
 - Ranking/visibility of minorities in networks
 - Link and content recommendation in networks
 - Psychological well-being
 - Upward social comparison
 - Loneliness and Facebook use: the role of social comparison and rumination (PMID: [33521361](#))